

13 Assumptions in Regression Analysis

In most practical regression problems, researchers have only sample data in their hand drawn from real or hypothetical population, and based on the results derived from sample data; they are supposed to give some decisions about the population. Therefore, it is important that the researchers understand and know the nature of the population which they are working with. They should know enough about the population to portrait a suitable model for the particular situation. However, defining each characteristic of a situation may produce a much complex model. Sometimes, they do not need to make the model perfect! But, then, how can a researcher isolate the occasions in which their chosen model and data are sufficiently comparable for them to proceed? Well; to ensure comparability (and to fulfill some other requirements) we set some assumptions.

The least squares fitting procedure can be used for data analysis as a purely descriptive technique. However, the procedure has strong theoretical justification if a few assumptions are made about how the data are generated. In the context of scientific investigation, "assumptions" are not just devices to simplify mathematics, they are supposed to be a reasonable mathematical representation of the data-generating process.

(a) Assumption 1: Linear Model

The population means of the values of the dependent variable Y at each value of the independent variable X are assumed to be on a straight line. That is, the Regression Model should be Linear in Parameters.

(b) Assumption 2: Non-random Value of Regressor

Values taken by the Regressor X are considered non-random / fixed in repeated sampling. This means our Regression Analysis is conditional on the given value of X. In an experiment, the values of the independent variable would be fixed by the experimenter and repeated samples could be drawn with the independent variables fixed at the same values in each sample. As a consequence of this assumption, the independent variables will in fact be independent of the disturbance.

(c) Assumption 3: Disturbance term with zero mean

The mean value of the random disturbance term is zero for the given value of X, that is, the conditional expected value of disturbance term with respect to X is zero. This assumption assures that the disturbance term, although makes individual Y values deviate from fitted line, but as a whole (while summed up), it poses no affect on the mean value of Y - by canceling out its positive values by its own negative values.

(d) Assumption 4: Disturbance term with constant variance

The conditional variance of Disturbance term with respect to given X values are same / identical / constant. In Statistics, we call this phenomenon of equal variance as HOMOSCEDASTICITY. In contrast, the condition of the error variance not being constant over all observations is called heteroscedasticity. This heteroscedasticity is inherent when the response in regression analysis follows a distribution in which the variance is functionally related to the mean or when there are several group with different variances.

Our current assumption simply states that whenever X varies, the corresponding Y populations have same variance around the Regression line. By imposing this assumption, we give emphasis to the fact the "All Y values corresponding to various X's are equally important."

If the "Reliability" being judged by how closely the Y values are distributed around their means; then all Y values corresponding to the various X's will not be equally reliable if this assumption does not hold good.

(e) Assumption 5: No Autocorrelation

For any two given X values the correlation between any two random disturbance is zero, which means no autocorrelation should exist. By this assumption, we make sure that we purely consider systematic effect of X on Y (if any exist) and we do not want to get worried about the influence that might act on Y as a result of possible inter-correlations among the e's simultaneously, because it

may make the situation worse while defining relationships between Y and X.

(f) Assumption 6: Stochastic Value of Regressor

If the X's are stochastic, the disturbance term and the random X's are independent or at least uncorrelated. This assumption is particularly important to make the model simple to calculate. If X and e are correlated, it is not possible to isolate / assess their individual influences on Y. This is why, we assume X and e have separate (additive in linear case) effects on Y.

Notice that, this assumption is not necessary if X is fixed (Assumption 2). The current assumption is a solution, only if the assumption 2 is violated.

(g) Assumption 7: Estimation Paradox

The number of observations, say n, must be greater than the number of "Parameters to be estimated", which in turns means- the number of sample values must be greater than the number of regressors.

(h) Assumption 8: Spread in Regressors

There must be sufficient variability in the values taken by the regressors.

This assumption 8 and the previous one, the assumption 7, are frequently overlooked by most text-book authors although these innocent looking assumptions are very hard to handle if they are violated. Violation of this

Assumption 8 makes it impossible to estimate regression parameters.

(i) Assumption 9: Model Specification

There should be no specification bias in the model used in empirical analysis, that is, the regression model should be correctly specified about

- Variables
- Functional form
- Assumption

The validity of the model is also a question of high importance.

However, in practice, investigator rarely knows the correct variable to include the model or the correct functional form of the model or the correct probabilistic assumptions about the variables entering the model. Therefore, there is some "Trial and Error" involved in choosing the right model for empirical situations. Then why we bother about this assumption at all if judgment is required in selecting a model?

Well, this is just a reminder of the fact that, since our entire regression analysis is based on the model - careful choice of model should be used in formulating models, especially when there are several competing theories.

(j) Assumption 10: Multicollinearity

There should be no exact or perfect linear relationships among the independent variables. We assume that the independent variables are linearly independent. That is, no independent variable can be expressed as a

(non-zero) linear combination of the remaining independent variables. The failure of this assumption, known as multicollinearity, clearly makes it infeasible to disentangle the effects of the supposedly independent variables and provide poor estimates of your regression coefficients. This assumption is particularly applicable in multiple regression containing several regressors.

(k) Assumption 11: Normality assumption

The Classical Normal Linear Regression model (CNLRM) specifically assumes that the stochastic disturbance term (which represent the combined influence on the dependent variable of a large number of independent variables that are not explicitly introduced in the regression model) is Normally distributed with- mean of disturbance term 0, variance constant, covariance between two disturbance term is zero. Here, we should note that, the only difference of Classical Normal Linear Regression model (CNLRM) with Classical Linear Regression model (CLRM) is that CLRM only requires that the mean is zero and the variance is a finite positive constant; no assumption or restriction about the probability distribution of is specified. The readers who are wondering why we are talking about Normal distribution in CNLRM, it is worth mentioning that the theoretical justification for this extra assumption is the Central Limit theorem itself.

Note that, normality of the predictors is not required in OLS, that is, there are no assumptions regarding the distribution of the independent variables in OLS.

(1) Assumption 12: Scale of measurement

Variables measured at interval or ratio levels. However, using other statistical techniques, we can fix the possible violation of this assumption.

(m) Assumption 13: Error free predictor

One of the assumptions of linear regression is that the independent variable is measured without error. Just a philosophical question ...

Is it possible to measure anything without error? However, there is a large literature on errors-in-variables regression which we like to skip here.

References

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